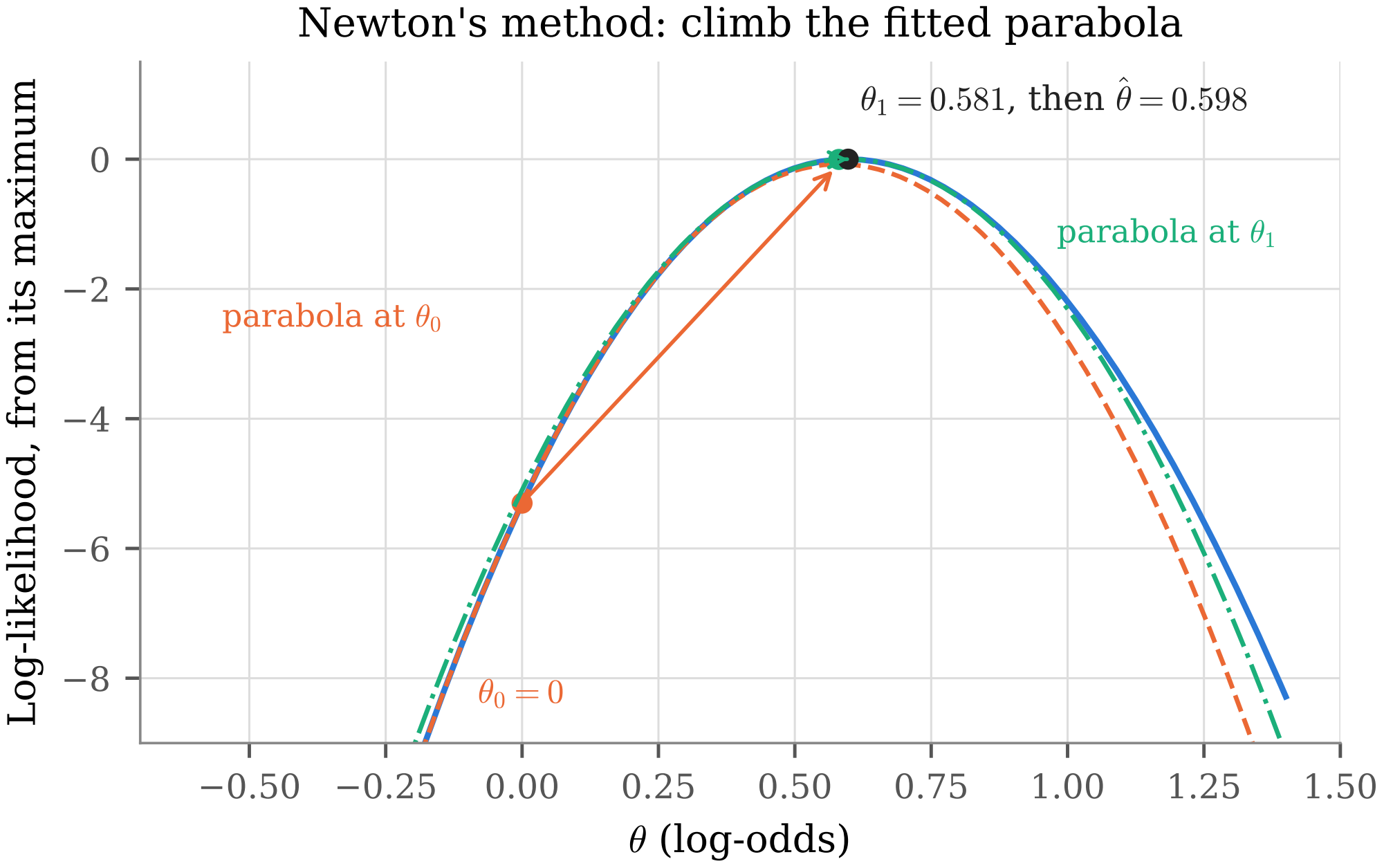
Newton's method: climb the fitted parabola
$\theta_1 = 0.581$, then $\hat{\theta} = 0.598$
parabola at θ_1
parabola at θ_0
$\theta_0 = 0$
Log-likelihood, from its maximum
0
−2
−4
−6
−8
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
1.50
θ (log-odds)